MILITARY SPECIFICATION

RADIAC CALIBRATOR SET AN/UDM-7( )

1. SCOPE

1.1 This specification covers a radiac calibrator set AN/UDM-7( ) for calibrating alpha radiac instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-1/1388 - Electron Tube, Type 7840.
MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-Q-9858 - Quality Program Requirements.
MIL-P-15328 - Primer Pretreatment (Formula No. 117 for Metals).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-M-19590 - Marking of Commodities and Containers to Indicate Radioactive Material.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibration of Shipboard Equipment.

DRAWINGS

MILITARY

RE101F2002 - Alpha Radiac Calibrator AN/UDM-7A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

INTERSTATE COMMERCE COMMISSION

Tariff No. 10 - Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles by Land and Water in Rail Freight Service and by Motor Vehicles (Highway and Water) including Specifications for Shipping Containers.

FSC 6665

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20360.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.2.1 (see 6.2).

3.2 General description.- The radiac calibrator set AN/UDM-7( ) is a calibrating set which provides suitable radioactive sources for calibrating alpha radiac survey equipment.

3.2.1 The calibration set shall be constructed and assembled in accordance with the requirements of this specification and Drawing RE101F2002. Where the requirements of this specification conflict with the drawing, the requirements of this specification shall govern.

3.3 Material.- Materials specified herein and in Drawing RE101F2002 shall be entirely suitable for the purpose intended. Use of other material shall have the approval of the procuring activity.

3.4 Equipment composition.- The calibration set shall consist of the following:

(a) Two radioactive sources "A" and "C".
(b) Two radioactive source holders.
(c) Adjustable positioner.
(d) Two attenuators.
(e) Aluminum housing.
(f) Calibrator carrying case.
(g) One pair of tweezers.
(h) Instruction book.

3.4.1 Two radioactive sources.- The two radioactive sources shall contain Plutonium-239 deposited in a resin component. The resin component shall be of the composition specified in MIL-P-15328.

3.4.1.1 Source configuration and description.- The sources shall be about 12-1/2 inches in diameter, 1.0 mg/cm^2 thick and emit alpha energies, 90 percent of which are 4MEV or greater. A set of two sources (labeled A and C, and of activities of 10^7 DPM and 10^5 DPM respectively) are contained in each unit. When a source is positioned in the unit, only a 4 inch x 10 inch area is exposed for calibration. The sources shall be prepared in accordance with the Appendix to this specification.

3.4.1.2 Uniformity of radioactive sources.- The sources shall have no area count rate which exceeds plus or minus 5 percent from the average count rate when tested as specified in 4.4.1.

3.4.1.3 Accuracy.- The activity of each source shall be determined by comparison with a standard source to be approved by the procuring activity. This information shall be recorded and included with each calibrator delivered under the contract. Information shall be on a printed card or similar method of presentation.

3.4.2 Source mounting.- The two radioactive sources shall be mounted in accordance with Drawing RE101F2002.

3.4.3 Adjustable probe positioner.- Construction of the probe positioner shall be in accordance with Drawing RE101F2002.

3.4.4 Aluminum housing.- Each calibration set shall be provided with a lightweight housing, made of aluminum in accordance with Drawing RE101F2002, designed to house the following:

(a) Two radioactive source holders.
(b) Accessory drawer.
(c) The top of the unit shall be designed as a source-positioning well.

The housing shall be labeled in accordance with MIL-M-19590.

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3.4.4.1 The two radioactive source holders shall be contained in removable drawers, numbered for source identification, and labeled in accordance with MIL-M-19590.

3.4.4.1.1 Each drawer with source shall be removable as a unit for replacement in the proper position under the probe positioner.

3.4.5 Calibrator carrying case.- The calibrator carrying case shall be constructed to contain the aluminum housing. The carrying case shall be constructed in accordance with Drawing RE101F2002.

3.4.5.1 Size and weight of calibrator.- The overall size and weight of the calibrator shall be as follows:

Height - 1-27/32 inches.
Width - 13-23/32 inches.
Depth - 13-17/32 inches.
Weight - Not to exceed 17 pounds.

3.4.5.2 Size and weight of carrying case.- The overall size and weight of the carrying case shall be as follows:

Height - 3-15/16 inches
Width - 14-7/8 inches
Depth - 15-1/16 inches
Weight - Not to exceed 7-1/2 pounds.

3.5 Shock and vibration.-

3.5.1 Shock.- Shock requirements shall be for grade A, class I, type A in accordance with MIL-S-901, except that the drop shall be 1, 2 and 3 feet in lieu of 1, 3 and 5 feet.

3.5.2 Vibration.- Vibration requirements shall be in accordance with type I of MIL-STD-167.

3.6 Degree of enclosure.- Degree of enclosure for the calibrator shall be splashproof in accordance with MIL-STD-108.

3.7 Temperature.- The sources shall be capable of withstanding a temperature of 120°F in an inverted position, and meet the requirements of 3.9.

3.8 Humidity.- The sources shall be capable of withstanding 95 percent relative humidity at 100°F for 4 hours, and meet the requirements of 3.9.

3.9 Leakage.- When tested in accordance with 4.4.5, the removable radioactive material shall not exceed 0.005 microcuries of Plutonium-239. Leakage test shall be performed immediately after the temperature, humidity, shock and vibration tests.

3.10 Workmanship.- The calibration set shall be manufactured and finished in a thoroughly workmanlike manner and shall be free from all burrs, rough edges, smudges and scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality control system.- The contractor shall provide and maintain a quality system acceptable to the government for the supplies covered by the contract. The system of quality control shall be in accordance with MIL-Q-9858. The procedures outlined in MIL-Q-9858 shall serve to supplement and implement the design, performance and test requirements of this specification.

4.2 General inspection.- The methods of examination and testing of the calibration sets shall fall within the following classifications:

(a) Preproduction inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 Preproduction inspection.- Preproduction inspection shall be made on the preproduction model and shall consist of the examination of 4.3 and the tests of 4.4.

4.2.2 Quality conformance inspection.- Quality conformance inspection shall consist of the production inspection of 4.2.2.1 and production control inspection of 4.2.2.2.

4.2.2.1 Production inspection.- Production inspection shall be made on each equipment offered for delivery to determine compliance with this specification. Production inspection shall consist of the examination of 4.3 and the uniformity, accuracy and leakage tests of 4.4.1, 4.4.2 and 4.4.5.

4.2.2.2 Production control inspection.- Production control inspection shall be made on one out of each 10 calibration sets produced, to be selected at random by the Government representative. Production control inspection shall consist of the examination of 4.3 and the tests specified in 4.4.

4.3 General examination.- The calibration set shall be examined to determine compliance with the requirements of this specification and shall include the following:

(a) Workmanship, assembly, size and fit.
(b) Materials, parts and finish.

4.4. Test procedures.-

4.4.1 Uniformity.- The uniformity check to determine conformance with 3.4.1.2 shall be made with a type 7840 tube conforming to MIL-E-1/1388, used with a conventional type scaler (1 megohm, 0.5 microsecond resolving time).

4.4.2 Accuracy.- The calibration set shall be tested to determine conformance with 3.4.1.3.

4.4.3 Enclosure.- The housing and the carrying case shall be tested to determine conformance with 3.6.

4.4.4 Shock and vibration.- The calibration set shall be tested to determine conformance with 3.5.1 and 3.5.2.

4.4.5 Leakage test.- The leakage test shall be capable of detecting the presence of 0.005 microcuries of Plutonium-239. The method of the test shall be submitted to the command or agency concerned for approval prior to performing the test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- The equipment shall be packaged by level A or C and packed by level A, B or C as specified in the contract or order in accordance with MIL-E-17555. Method III preservation shall apply for level A packaging. As a minimum, the requirements shall conform to the Interstate Commerce Commission Tariff No. 10.

5.2 Marking.- The equipment and containers shall be marked in accordance with MIL-M-19590.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-SH
(Project 6665-N228Sh)

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APPENDIX

PROCEDURE FOR PREPARING "THIN" ALPHA SOURCES

10. Technique.- The technique is relatively simple. The alpha emitter is dissolved in an alcohol-resin component mixture and stirred to uniformity. The final mixture is then poured through a funnel onto a plastic disk located on a leveling table directly below the funnel spout. The resulting radioactive sources have excellent uniformities - with any source the source area count rate varies ±5 percent or less from the average count rate.

10.1 Procedure.- The procedure shall be as follows:

(a) One cubic centimeter (cc) of radioactive solution (isotope in chloride form) is added to a mixture of 9 - 10 cc of resin component (MIL-C-15328) and 29 cc of ethyl alcohol (99 percent). The total mixture is carefully stirred for a minimum of 15 minutes to assure a uniform dispersion of the radioisotope in the solution.

(b) The radioactive solution is then poured through a funnel onto a 12-1/2 ($\pm^{0}_{1/64}$) inch diameter disk on a leveled table. The funnel (spout inner diameter - 11 millimeter (mm) is positioned rigidly with the spout perpendicular to the center of the plastic disk (CR-39 transparent plastic; cast acrylic). The distance between spout and disk is 47 mm. The spout inner diameter and the 47 mm distance eliminate areas of reduced activity in the center of the disk. Prior to pouring, the disk is leveled by means of a leveling table with adjustable legs (NASL uses a 12 inches x 12 inches stainless steel table). A wetting agent applied to the surface of the disk before pouring facilitates spreading of the radioactive mixture. Ten cc of alcohol, carefully hand spread over the disk has bee been used with good results.

(c) After spreading freely on the disk, the radioactive mixture is allowed to air dry. During this phase two factors of control are necessary. First, safety precautions are needed to protect personnel and equipment from possible contamination. A hood is recommended, with conditions to keep air currents from passing over the drying radioactive liquid. Secondly, humidity control is important and a dust free atmosphere is desirable. A relative humidity of 40 percent or less at 70°F. or less will prevent spotty, nonuniform distributions. The water content of the mixture is critical and should never be allowed to exceed 0.3 cc in 5 cc of mixture.

10.2 Source.- The resulting source, if the conditions above are employed, will have a high degree of uniformity. Any area count rate will be ±5 percent or less from the average count rate. A uniformity check can be made with 7840 GM tube (mica window thickness of 2.5 mg/cm^2 or less) used with a conventional type scaler (1 megohm, 0.5 μsec resolving time). The alpha source is a "thin" source, meaning all alpha energies at the surface are 4 Mev or greater. Care should be taken not to gouge or severely rub the source material.

10.3 Summary of material and conditions.- Summary of material and conditions shall be as follows:

(a) Radioactive mixture.-

Radioisotope - Plutonium 239 as $PuCl_3$ in HCl (≤ 0.5 N)
810 μg/cc for source 1 - equivalent to approximately 10^4 μg/m^2
8.1 μg/cc for source 2 - equivalent to approximately 10^2 μg/m^2
Resin component - 9 - 10 cc of MIL-C-15328
Solvent - 29 cc of ethyl alcohol (99 percent)
Total volume
of mixture - 40 cc - bring up with alcohol if necessary

(b) Plastic disk - 12-1/2 inch diameter, 1/8 inch thick, CR-39 transparent, cast acrylic
(c) Leveling table - At least 12 inches x 12 inches for relatively uniform evaporation
(d) Funnel - Spout (inner diameter - 11mm, length - 25mm)
Mouth diameter ≈65mm
(e) Mixture beaker - 125 cc (graduated) with pouring spout

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(f) Stirring rod - Glass, durable, approximately 8 inches long
(g) Conditions - R. H. - 40 percent or less, temperature 70° F or less
Dust-free atmosphere
Hood for safety precautions
No wind currents over drying mixture

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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